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22ESC143/243

VTU.REG.NO

B. E. First / Second Semester Examination – November 2024

(Summer Term)

Introduction to Electronics Engineering

Time: 3 hrs] [Maximum Marks: 100

Note: Answer any FIVE full questions, selecting at least ONE full question from each Module.

Module – I

1. a) What do you mean by regulated power supply? Explain the same with the help of general block schematic. **(06 Marks)**

   b) With the aid of neat diagram and waveforms explain the working of a bridge rectifier. Also mention the advantage of bridge rectifier over two-diode full wave rectifier. **(09 Marks)**

   c) Derive an expression for the Ripple factor of full wave rectifier. Also mention its PIV. **(05 Marks)**

2. a) What is a voltage Multiplier? Explain voltage Doublers and Triplers with the help of neat diagrams. **(08 Marks)**

   b) List the need for negative feedback in amplifier. Also draw the circuit diagrams of a typical 2-stage R-C coupled amplifier in CE configuration and mention the roles of emitter by pass capacitor and a coupling capacitor. **(08 Marks)**

   c) A Full wave rectifier has a load of 1kΩ. Applied input AC voltage is 200-0-200 Volts. Compute i) DC current ii) Ripple factor iii) DC output power. **(04 Marks)**

Module - II

3. a) What is operational amplifier? Define the following terms with respect to OP AMP: Common mode gain, CMRR and slew rate and differential gain. **(06 Marks)**

   b) Explain the use of OP AMP as integrators, differentiator and adder with the help of neat sketches and the relevant equations. **(08 Marks)**

   c) Explain the working of Astable Multivibrator with the help of neat diagram and relevant waveforms **(06 Marks)**

4. a) State and explain Barkhausen criterion for sustained oscillations. Also explain the role of ladder network in oscillators. **(07 Marks)**

   b) Explain the working of RC phase shift oscillator using OP AMP with the help of neat circuit diagram. Also mention the relevant equations as applied to this oscillator. **(09 Marks)**

   c) Compute the value of capacitor for output frequency of 1 kHz in RC phase shift oscillator. It is given that R = 100 kΩ **(04 Marks)**

Module – III

5. a) Perform the following conversions: i) $(126.25)_{10}$ to binary
   ii) $(11110101011101.111010)_2$ to octal and hexadecimal
   iii) $(10011011.110)_2$ to decimal iv) $(FCD.AC)_{16}$ to decimal. **(08 Marks)**

   b) Perform subtraction using 2's compliment method
   i) $(11010)_2 - (10000)_2$ ii) $(100)_2 - (110000)_2$. **(04 Marks)**

   c) Justify the statement: "NAND and NOR gates are universal gates" with the help of relevant illustrations. **(08 Marks)**

6. a) Explain the following : Canonical minimum and maximum term forms, SOP and POS forms with the aid of suitable illustrations. **(09 Marks)**

b) What are Half and Full Adders? Explain their operation with the help of truth tables and relevant Boolean expressions. **(11 Marks)**

Module - IV

7. a) What are embedded systems? Give examples their classifications, major applications and their elements. **(10 Marks)**

b) In the context of embedded systems, briefly explain the following: Transducers sensors, Actuators, ES cores, LEDs and Led displays. **(10 Marks)**

8. a) Give the difference between general purpose computing systems and embedded systems. **(08 Marks)**

b) Explain and make the distinction between the following: i) Microprocessor and Microcontroller ii) RISC and CISC iii) Havvard and Von-Neumann architectures. **(12 Marks)**

Module - V

9. a) Explain the need for communication. Also explain the general form of communication system with the help of a neat block diagram. **(11 Marks)**

b) Explain the following terms: Noise, signal to noise ratio, noise figure, multiplexing. **(09 Marks)**

10. a) With the help of relevant waveforms, explain AM, FM and FSK. **(09 Marks)**

b) Explain the process of ASK and PSK with the help of neat waveforms. **(05 Marks)**

c) Write short notes on multiple Access Techniques and Radio wave propagation. **(06 Marks)**